

SHETH LUJ AND SIR MV COLLEGE
Cyber & Information Security

Practical No 1

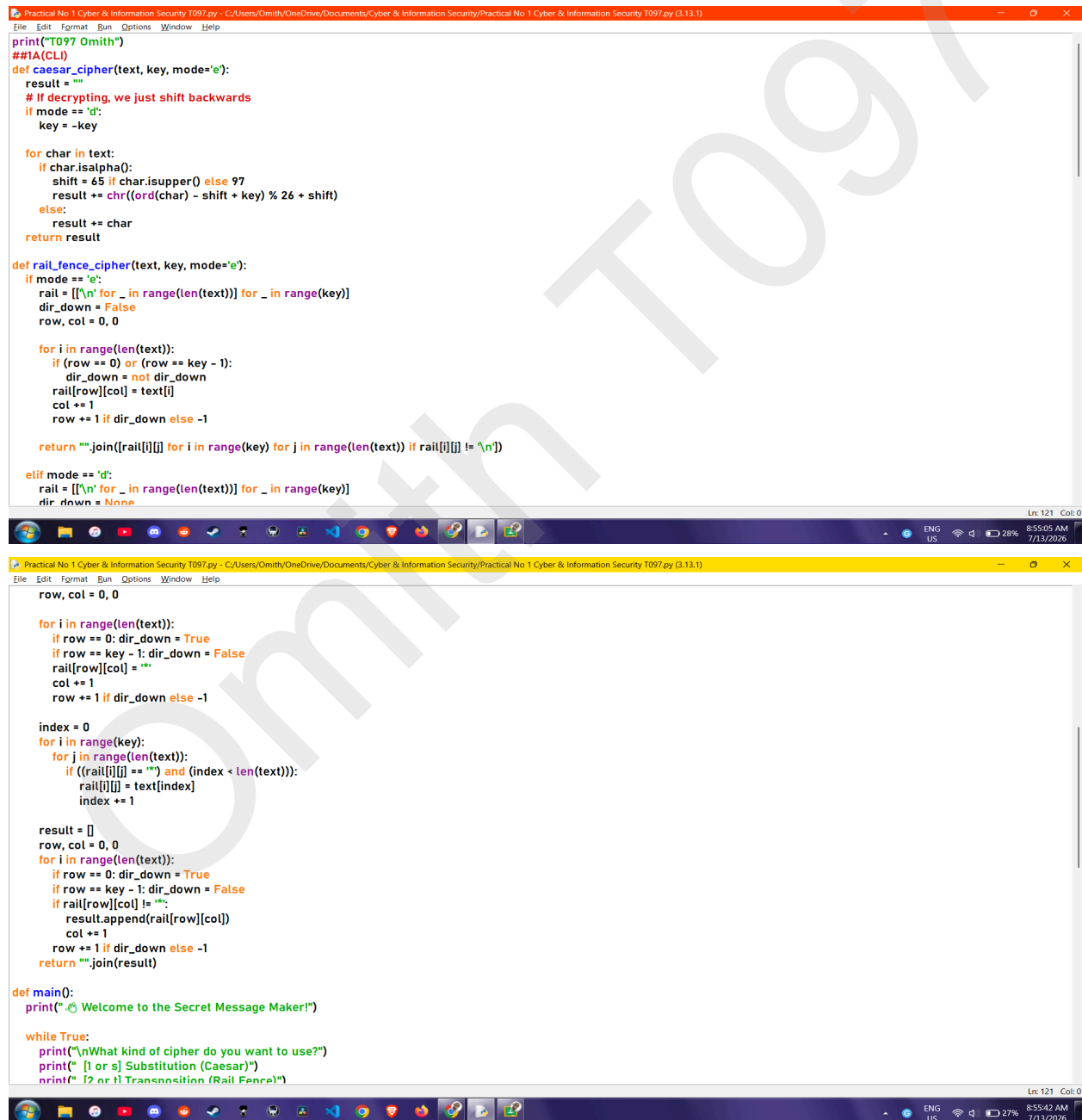
Aim:

1A) Design and implement algorithms to encrypt and decrypt messages using classical substitution techniques

1B) Design and implement algorithms to encrypt and decrypt messages using transposition techniques

1A. CLI Version:

Code:



```
Practical No 1 Cyber & Information Security T097.py - C:/Users/Omith/OneDrive/Documents/Cyber & Information Security/Practical No 1 Cyber & Information Security T097.py (3.13.1)
File Edit Format Run Options Window Help
print("T097 Omith")
##1A(CLI)
def caesar_cipher(text, key, mode='e'):
    result = ""
    # If decrypting, we just shift backwards
    if mode == 'd':
        key = -key
    for char in text:
        if char.isalpha():
            shift = 65 if char.isupper() else 97
            result += chr((ord(char) - shift + key) % 26 + shift)
        else:
            result += char
    return result

def rail_fence_cipher(text, key, mode='e'):
    if mode == 'e':
        rail = [['\n' for _ in range(len(text))] for _ in range(key)]
        dir_down = False
        row, col = 0, 0
        for i in range(len(text)):
            if (row == 0) or (row == key - 1):
                dir_down = not dir_down
            rail[row][col] = text[i]
            col += 1
            row += 1 if dir_down else -1
        return "".join([rail[i][j] for i in range(key) for j in range(len(text)) if rail[i][j] != '\n'])
    elif mode == 'd':
        rail = [['\n' for _ in range(len(text))] for _ in range(key)]
        dir_down = None
        row, col = 0, 0
        for i in range(len(text)):
            if row == 0: dir_down = True
            if row == key - 1: dir_down = False
            rail[row][col] = ""
            col += 1
            row += 1 if dir_down else -1
        index = 0
        for i in range(key):
            for j in range(len(text)):
                if ((rail[i][j] == "") and (index < len(text))):
                    rail[i][j] = text[index]
                    index += 1
        result = []
        row, col = 0, 0
        for i in range(len(text)):
            if row == 0: dir_down = True
            if row == key - 1: dir_down = False
            if rail[row][col] != "":
                result.append(rail[row][col])
                col += 1
            row += 1 if dir_down else -1
        return "".join(result)

def main():
    print("👋 Welcome to the Secret Message Maker!")
    while True:
        print("\nWhat kind of cipher do you want to use?")
        print("  [1 or s] Substitution (Caesar)")
        print("  [2 or t] Transposition (Rail Fence)")
```

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print(" [2 or t] Transposition (Rail Fence)")
print(" [q] Quit")

choice = input("Your choice: ").strip().lower()

if choice == 'q':
    print("Catch you later! Stay secure. 🛡️ 📧")
    break

if choice not in ['1', 's', '2', 't']:
    print("Oops, I didn't recognize that choice. Let's try again.")
    continue

text = input("\nWhat's your message? ")

# Friendly error handling for the key
try:
    key = int(input("Enter your secret key (a whole number): "))
except ValueError:
    print("Hmm, that doesn't look like a number. Let's start over!")
    continue

mode_input = input("Do you want to (e)ncrypt or (d)ecrypt? ").strip().lower()

# Map 'encrypt' or 'e' to just 'e'
if mode_input.startswith('e'):
    mode = 'e'
    action_word = "Encrypted"
elif mode_input.startswith('d'):
    mode = 'd'
    action_word = "Decrypted"
else:
    print("Please type 'e' or 'd'. Let's start over!")
    continue

mode_input = input("Do you want to (e)ncrypt or (d)ecrypt? ").strip().lower()

# Map 'encrypt' or 'e' to just 'e'
if mode_input.startswith('e'):
    mode = 'e'
    action_word = "Encrypted"
elif mode_input.startswith('d'):
    mode = 'd'
    action_word = "Decrypted"
else:
    print("Please type 'e' or 'd'. Let's start over!")
    continue

# Execute the chosen cipher
if choice in ['1', 's']:
    output = caesar_cipher(text, key, mode)
else:
    if key < 2:
        print("For Rail Fence, the key needs to be 2 or higher!")
        continue
    output = rail_fence_cipher(text, key, mode)

print("\n" * 30)
print(f"🌟 Here is your {action_word} message:")
print(f"📄 {output}")
print("\n" * 30)

if __name__ == "__main__":
    main()

##1A(GUI)
```

Output:

Omith Thilakan
T097

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```
File Edit Shell Debug Options Window Help
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/Omith/OneDrive/Documents/Cyber & Information Security/Practical No 1 Cyber & Information Security T097.py
T097 Omith
Welcome to the Secret Message Maker!

What kind of cipher do you want to use?
[1 or s] Substitution (Caesar)
[2 or t] Transposition (Rail Fence)
[q] Quit
Your choice: 1

What's your message? Hello
Enter your secret key (a whole number): 3
Do you want to (e)ncrypt or (d)ecrypt? e
-----
Here is your Encrypted message:
Khoor
-----

What kind of cipher do you want to use?
[1 or s] Substitution (Caesar)
[2 or t] Transposition (Rail Fence)
[q] Quit
Your choice: 1

What's your message? Khoor
Enter your secret key (a whole number): 3
Do you want to (e)ncrypt or (d)ecrypt? d
-----
Here is your Decrypted message:
Hello
-----

Ln: 40 Col: 13

File Edit Shell Debug Options Window Help
What kind of cipher do you want to use?
[1 or s] Substitution (Caesar)
[2 or t] Transposition (Rail Fence)
[q] Quit
Your choice: 2

What's your message? Hello
Enter your secret key (a whole number): 3
Do you want to (e)ncrypt or (d)ecrypt? e
-----
Here is your Encrypted message:
Hoell
-----

What kind of cipher do you want to use?
[1 or s] Substitution (Caesar)
[2 or t] Transposition (Rail Fence)
[q] Quit
Your choice: 2

What's your message? Hoell
Enter your secret key (a whole number): 3
Do you want to (e)ncrypt or (d)ecrypt? d
-----
Here is your Decrypted message:
Hello
-----

What kind of cipher do you want to use?
[1 or s] Substitution (Caesar)
[2 or t] Transposition (Rail Fence)
[q] Quit
Your choice:
Ln: 68 Col: 13
```

1A. GUI Version:

Code:

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T097

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```
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"""A(GUI)"""
import tkinter as tk
from tkinter import ttk, messagebox

# --- Core Cryptography Logic ---

def caesar_cipher(text, key, mode='e'):
    result = ""
    if mode == 'd':
        key = -key
    for char in text:
        if char.isalpha():
            shift = 65 if char.isupper() else 97
            result += chr((ord(char) - shift + key) % 26 + shift)
        else:
            result += char
    return result

def rail_fence_cipher(text, key, mode='e'):
    if mode == 'e':
        rail = [['\n' for _ in range(len(text))] for _ in range(key)]
        dir_down = False
        row, col = 0, 0
        for i in range(len(text)):
            if (row == 0) or (row == key - 1): dir_down = not dir_down
            rail[row][col] = text[i]
            col += 1
            row += 1 if dir_down else -1
        return "".join([rail[i][j] for i in range(key) for j in range(len(text)) if rail[i][j] != '\n'])
    elif mode == 'd':
        rail = [['\n' for _ in range(len(text))] for _ in range(key)]
        dir_down = None
        row, col = 0, 0

row, col = 0, 0
for i in range(len(text)):
    if row == 0: dir_down = True
    if row == key - 1: dir_down = False
    rail[row][col] = ""
    col += 1
    row += 1 if dir_down else -1
index = 0
for i in range(key):
    for j in range(len(text)):
        if rail[i][j] == "" and index < len(text):
            rail[i][j] = text[index]
            index += 1
result = []
row, col = 0, 0
for i in range(len(text)):
    if row == 0: dir_down = True
    if row == key - 1: dir_down = False
    if rail[row][col] != "":
        result.append(rail[row][col])
        col += 1
    row += 1 if dir_down else -1
return "".join(result)

# --- GUI Application Logic ---

def process_text():
    text = text_entry.get("1.0", "end-1c")
    if not text.strip():
        messagebox.showwarning("Empty Input", "Please enter a message to process.")
        return
    try:
        key = int(key_entry.get())
```

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except ValueError:
    messagebox.showerror("Invalid Key", "Your secret key must be a whole number.")
    return

mode = mode_var.get()
cipher_type = cipher_var.get()

if cipher_type == "s":
    output = caesar_cipher(text, key, mode)
else:
    if key < 2:
        messagebox.showerror("Invalid Key", "Rail Fence requires a key of 2 or more.")
        return
    output = rail_fence_cipher(text, key, mode)

# Animate output update
output_entry.config(state="normal")
output_entry.delete("1.0", "end")
output_entry.insert("1.0", output)
output_entry.config(state="disabled")

def copy_to_clipboard():
    output = output_entry.get("1.0", "end-1c")
    if output.strip():
        root.clipboard_clear()
        root.clipboard_append(output)
        messagebox.showinfo("Success", "Copied to clipboard! 📋")
    else:
        messagebox.showwarning("Empty", "Nothing to copy yet.")

def clear_all():
    text_entry.delete("1.0", "end")
    key_entry.delete(0, "end")
    output_entry.config(state="normal")

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# --- UI Setup & Layout ---

# Define Colors for Dark Theme
BG_COLOR = "#1e1e2e"
FG_COLOR = "#cdd6f4"
ACCENT_COLOR = "#a6e3a1"
HOVER_COLOR = "#94acc9"
ENTRY_BG = "#313244"

root = tk.Tk()
root.title("Advanced Cryptography Tool")
root.geometry("550x680")
root.configure(bg=BG_COLOR)
root.resizable(False, False)

# Styling using ttk
style = ttk.Style()
style.theme_use('clam')

# Configure Radiobutton style for dark theme
style.configure("Dark.TRadiobutton", background=BG_COLOR, foreground=FG_COLOR, font=("Segoe UI", 10))
style.map("Dark.TRadiobutton",
    background=[('active', BG_COLOR)],
    indicatorcolor=[('selected', ACCENT_COLOR), ('!selected', ENTRY_BG)])

# Title Header
header = tk.Label(root, text="🔒 Secret Message Maker", font=("Segoe UI", 18, "bold"), bg=BG_COLOR, fg=ACCENT_COLOR)
header.pack(pady=(20, 15))

# Main Container
frame = tk.Frame(root, bg=BG_COLOR)
frame.pack(padx=30, fill="both", expand=True)
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tk.Label(frame, text="1. Enter Your Message:", font=("Segoe UI", 11, "bold"), bg=BG_COLOR, fg=FG_COLOR).pack(anchor="w")
text_entry = tk.Text(frame, height=4, font=("Consolas", 11), bg=ENTRY_BG, fg=FG_COLOR, insertbackground=FG_COLOR, relief="flat", padx=10, pady=10)
text_entry.pack(fill="x", pady=(5, 15))

# Key Input
tk.Label(frame, text="2. Secret Key (Integer):", font=("Segoe UI", 11, "bold"), bg=BG_COLOR, fg=FG_COLOR).pack(anchor="w")
key_entry = tk.Entry(frame, font=("Consolas", 12), bg=ENTRY_BG, fg=FG_COLOR, insertbackground=FG_COLOR, relief="flat")
key_entry.pack(fill="x", pady=(5, 15), ipady=5)

# Options Frame (Side-by-side Layout)
options_frame = tk.Frame(frame, bg=BG_COLOR)
options_frame.pack(fill="x", pady=10)

# Cipher Selection Column
cipher_frame = tk.Frame(options_frame, bg=BG_COLOR)
cipher_frame.pack(side="left", expand=True, fill="both")
tk.Label(cipher_frame, text="3. Choose Cipher:", font=("Segoe UI", 11, "bold"), bg=BG_COLOR, fg=FG_COLOR).pack(anchor="w", pady=(0, 5))

cipher_var = tk.StringVar(value="s")
ttk.Radiobutton(cipher_frame, text="Caesar (Substitution)", variable=cipher_var, value="s", style="Dark.TRadiobutton").pack(anchor="w", pady=2)
ttk.Radiobutton(cipher_frame, text="Rail Fence (Transposition)", variable=cipher_var, value="t", style="Dark.TRadiobutton").pack(anchor="w", pady=2)

# Mode Selection Column
mode_frame = tk.Frame(options_frame, bg=BG_COLOR)
mode_frame.pack(side="right", expand=True, fill="both")
tk.Label(mode_frame, text="4. Choose Action:", font=("Segoe UI", 11, "bold"), bg=BG_COLOR, fg=FG_COLOR).pack(anchor="w", pady=(0, 5))

mode_var = tk.StringVar(value="e")
ttk.Radiobutton(mode_frame, text="Encrypt Message", variable=mode_var, value="e", style="Dark.TRadiobutton").pack(anchor="w", pady=2)
ttk.Radiobutton(mode_frame, text="Decrypt Message", variable=mode_var, value="d", style="Dark.TRadiobutton").pack(anchor="w", pady=2)

# Buttons Frame
btn_frame = tk.Frame(frame, bg=BG_COLOR)
btn_frame.pack(fill="x", pady=20)

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File Edit Format Run Options Window Help
ttk.Radiobutton(mode_frame, text="Encrypt Message", variable=mode_var, value="e", style="Dark.TRadiobutton").pack(anchor="w", pady=2)
ttk.Radiobutton(mode_frame, text="Decrypt Message", variable=mode_var, value="d", style="Dark.TRadiobutton").pack(anchor="w", pady=2)

# Buttons Frame
btn_frame = tk.Frame(frame, bg=BG_COLOR)
btn_frame.pack(fill="x", pady=20)

# Custom Hover Function for Main Button
def on_enter(e):
    magic_btn.config(bg=HOVER_COLOR)
def on_leave(e):
    magic_btn.config(bg=ACCENT_COLOR)

clear_btn = tk.Button(btn_frame, text="Clear All", command=clear_all, font=("Segoe UI", 10, "bold"), bg="#f38ba8", fg="#1e1e2e", relief="flat", cursor="hand2", padx=15, pady=8)
clear_btn.pack(side="left")

magic_btn = tk.Button(btn_frame, text="🔮 Process Message", command=process_text, font=("Segoe UI", 12, "bold"), bg=ACCENT_COLOR, fg="#1e1e2e", relief="flat", cursor="hand2", padx=15, pady=8)
magic_btn.pack(side="right")
magic_btn.bind("<Enter>", on_enter)
magic_btn.bind("<Leave>", on_leave)

# Output Section
output_header = tk.Frame(frame, bg=BG_COLOR)
output_header.pack(fill="x")
tk.Label(output_header, text="Result:", font=("Segoe UI", 11, "bold"), bg=BG_COLOR, fg=FG_COLOR).pack(side="left")
tk.Button(output_header, text="📄 Copy", command=copy_to_clipboard, font=("Segoe UI", 8), bg=ENTRY_BG, fg=FG_COLOR, relief="flat", cursor="hand2").pack(side="right")

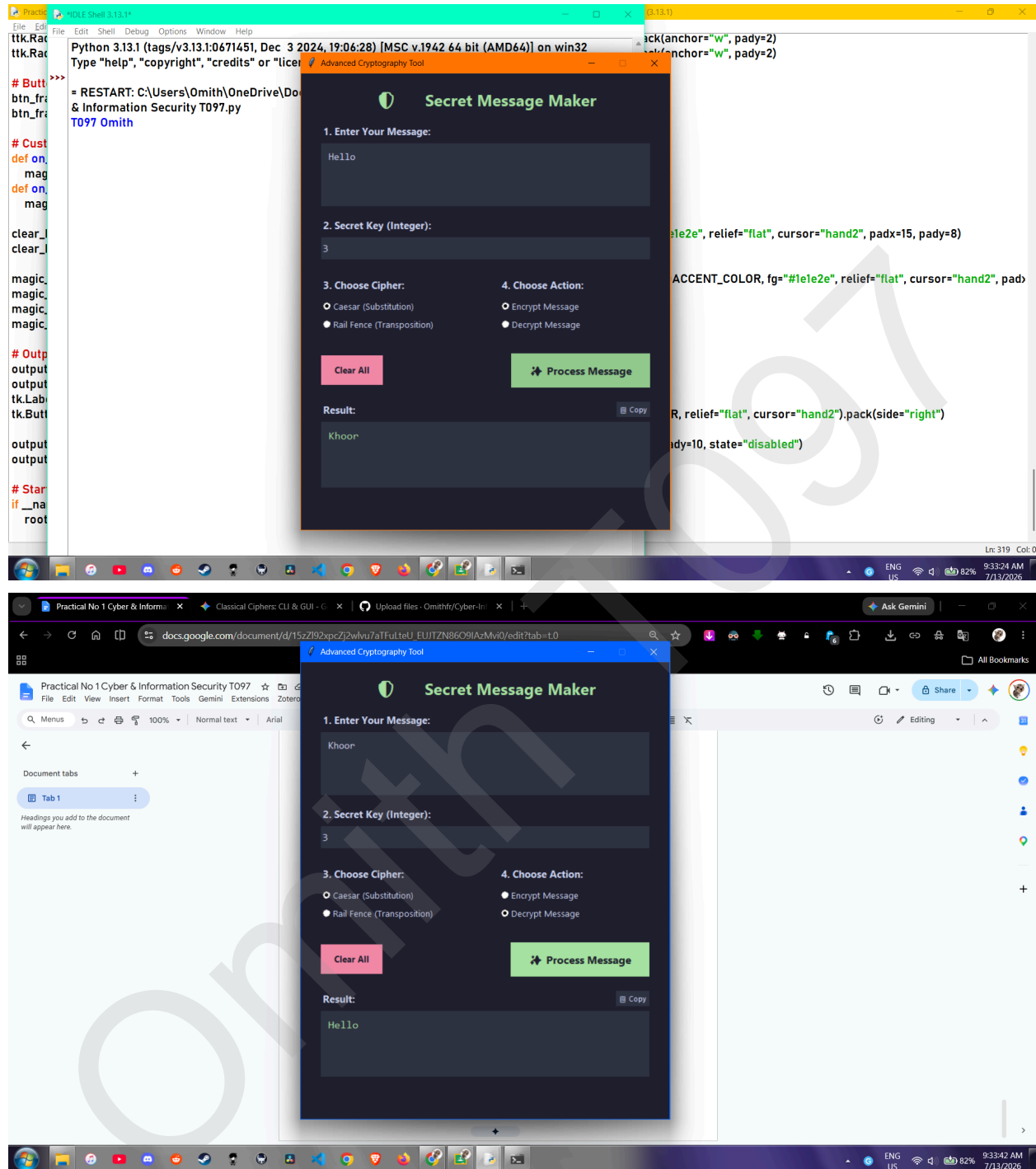
output_entry = tk.Text(frame, height=4, font=("Consolas", 12), bg=ENTRY_BG, fg=ACCENT_COLOR, relief="flat", padx=10, pady=10, state="disabled")
output_entry.pack(fill="x", pady=(5, 10))

# Start the application loop
if __name__ == "__main__":
    root.mainloop()
```

Output:

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